The S^6 Complex-Threefold Construction

A two-page mathematical status after proof reconstruction

Dated corrective audit — 25 August 2026

Current disposition. The earlier audit's topology rejection is withdrawn as stated. The manuscript contains two mathematically related specialization routes, and the first audit treated a bridge needed by the explicit collapse route as though it were needed by both. The non-collapse route through intrinsic nearby cycles, Leray transgression, and based van Kampen survives the present line-by-line reconstruction. The resulting review is provisionally favorable, with focused checks still requested for the global analytic construction and the external theorems invoked in nearby-cycle, Gysin, duality, and base-change arguments.

Construction and arithmetic datum. The manuscript constructs a compact complex threefold

$$f : X \longrightarrow \mathbb{P}^1$$

with generic fibre a two-dimensional complex torus, one reduced non-normal normal-crossings fibre W, and two multiple fibres of multiplicities 3 and 4. On the rank-four lattice, the exact monodromy calculation gives

$$T_1^3 = T_2^4 = I, \qquad T_0 = (T_1 T_2)^{-1} = I + N, \qquad N^2 = 0, \quad \operatorname{rk} N = 2.$$

The logarithmic twists satisfy

$$(\ell_0, \ell_1, \ell_2) = (0, 1, -1), \qquad p = 12\ell_0 - 4\ell_1 - 3\ell_2 = -1.$$

That same integer controls both the fundamental-group presentation and the three Leray transgressions.

The morphism-level correction. The explicit cusp-collapse map $c : F \to W$ supplies

$$c^* : H^q(W; \mathbb{Z}) \longrightarrow H^q(F; \mathbb{Z}),$$

whereas the nearby-cycle construction supplies the intrinsic stalk map

$$\mathrm{sp}^q : H^q(W; \mathbb{Z}) \longrightarrow H^q(F; \mathbb{Z})^{T_0}.$$

The manuscript identifies them when the collapse construction is used, but Section 7.7 requires only sp^q. Appendix B proves this map independently of c by the local models $t = z_0$, $t = z_0 z_1$, $t = z_0 z_1 z_2$, the associated nearby-cycle sheaves, the conductor/normalization calculation on W, hypercohomology rank bookkeeping, a degree-three support/Gysin argument, and integral saturation. The resulting maps are

$$\mathrm{sp}^q : H^q(W; \mathbb{Z}) \xrightarrow{\sim} H^q(F; \mathbb{Z})^{T_0}, \qquad 0 \le q \le 4,$$

with invariant ranks $(1, 2, 4, 2, 1)$, independently reproduced by exact exterior-power and Smith-normal-form calculations.

Leray and fundamental group. The central stalk above and the finite-fibre indices give

$$H^0(B, R^1 f_* \mathbb{Z}) = 12\mathbb{Z}\gamma, \quad H^0(B, R^2 f_* \mathbb{Z}) = 2\mathbb{Z}q, \quad H^0(B, R^3 f_* \mathbb{Z}) = 2\mathbb{Z}\gamma uw.$$

Local lift discrepancies around the three based meridians produce

$$d_2^{0,1}(12\gamma) = \pm p\omega,$$

and multiplicativity gives, with the same sign,

$$d_2^{0,2}(2q) = \pm p\,\omega\bar{\delta}, \qquad d_2^{0,3}(2\gamma uw) = \pm p\,\omega[\gamma\delta].$$

Since $p = -1$, all three maps are units, so $H^1(X; \mathbb{Z}) = H^2(X; \mathbb{Z}) = H^3(X; \mathbb{Z}) = 0$; duality and universal coefficients give the integral homology of S^6.

The based overlap calculation uses explicit deck transformations and gives

$$\pi_1(X) = \langle c, x, y \mid c \text{ central},\ x^3 = c^{\ell_1},\ y^4 = c^{\ell_2},\ xy = c^{\ell_0} \rangle.$$

Its relation determinant is $-p$, its actual Smith form is $\mathrm{diag}(1, 1, 1)$, and the nonabelian elimination yields a cyclic group of order $|p|$; hence $\pi_1(X) = 1$. Hurewicz, Whitehead, and $\Theta_6 = 0$ then identify the underlying smooth manifold with S^6.

Independent algebraic-dimension result. For an integral bivector E, the fibrewise $(1,1)$ condition is the vanishing of

$$P_E = E_{\gamma u} - E_{\gamma w}\tau - E_{\gamma\delta}\mu + 6E_{uw}\mu + E_{u\delta}\beta + E_{w\delta}(6\mu^2 - \tau\beta).$$

The transformation laws under the two generators prove linear independence of

$$1,\ \tau,\ \mu,\ \beta,\ 6\mu^2 - \tau\beta.$$

Thus the generic integral kernel is generated by

$$\eta = u \wedge w + 6\gamma \wedge \delta, \qquad \eta^2 = 12.$$

The associated Hermitian form has determinant $\frac{2}{3}\operatorname{Im}\tau\, D < 0$ and signature $(1,1)$. The generic fibre therefore has Néron–Severi group $\mathbb{Z}\eta$, while the torus algebraic-reduction alternatives of dimensions 1 and 2 are excluded. A meromorphic function on X is constant along a very general fibre because the analytic locus where its vertical differential vanishes contains those fibres and hence all of X. It descends to the base, giving

$$\mathcal{M}(X) = f^*\mathbb{C}(t), \qquad a(X) = 1,$$

and f is the algebraic reduction. This proof does not use Section 7 or the CDP comparison.

Section 10 mechanism. For a reduced normal-crossings non-normal fibre, the normalization/conductor sequence is exact in the local models $xy = 0$ and $xyz = 0$. The ambient differential of the defining equation has coefficients

$$(y, x, 0) \quad \text{or} \quad (yz, xz, xy),$$

so it vanishes on the conductor and glues to a nonzero ambient one-form whose intrinsic image is conductor-supported torsion. This verifies the local mechanism by which the ambient differential can evade a conormal reduction. Its global deployment uses the previously established topology and Picard information together with analytic Serre–Grothendieck duality and top-degree base change; those external inputs remain explicit review targets.

What remains to be checked.

- The explicit cusp-collapse map and the first Mayer–Vietoris route remain open as separate claims: continuity across every toric stratum, quotient descent, and the resulting integral collapse matrices have not been fully rederived at chain level. They are redundant for the non-collapse recognition route but still matter for the manuscript's claimed first topology proof and geometric description of W.
- Sections 3–6 have been read for coherence, and their finite transformation/freeness subclaims pass exact checks. A publication-level audit should still verify every proper-discontinuity, Hausdorffness, compactness, and holomorphic-gluing hypothesis lemma by lemma.
- Appendix B and Section 10 use standard external formalism. The next high-value task is a theorem-by-theorem source audit recording the exact hypotheses, naturality statements, signs, and integral saturation conclusions used here.

Provenance. The candidate manuscript supplies the original construction. The Product/Codex workbench identified the independent Leray and direct meromorphic-descent routes and repaired two transcription-level mistakes in its own reconstruction. The present integration treated that progress record as a roadmap, returned to the canonical 108-page source, reproduced the finite algebra, and reclassified each route separately. No claim here relies on inaccessible live workbench files.

Evidence pointers: revised proof audit; revised theorem and risk ledgers; 17-check integration harness; prior 38-check core harness; revised 108-page annotation overlay.